

Primitive Registry Ladder and Frame-Imposition Synthesis

v0.3 - Canon-adjacent protocol module for DST / METALABE primitive registry application

Status	ACTIVE-PROVISIONAL / HELD-TOWARD-LOAD-BEARING
Claim type	Semantic registry protocol / primitive construction grammar / PvNP modeling specification
Formal status	Not a proof of P vs NP or replacement for formal mathematics; a translation and audit layer
Core maxim	The square does not found the ladder. The ladder constructs the square.

Abstract

This document formalizes the primitive registry synthesis reached after revising the square, triangle, and circle grid framings. Its central claim is that a primitive registry system is a map-making choice that determines what kinds of relation, resonance, closure, transfer, evidence, and residue become legible. A registry does not merely relabel content. It changes the grammar of admissible evidence and therefore changes what appears locally compressible, globally stable, or difficult to derive.

The framing of consideration is imposed by an observer, instrument, organism, culture, or formal system as a registry choice. If a registry is constructed, it is still constructed from an imposed frame of consideration. This creates an infinite regress if treated as final ontology. DST therefore treats each registry as a fidelity contract within a declared frame, not as universal accounting with global fidelity.

The protocol begins from one marked referent and one degree of freedom. Mature structures such as horizontal/vertical, x/y, positive/negative, inside/outside, quadrants, and coordinate grids are downstream constructions, not primitives. The ladder builds through expression, trace, return, chirality, closure, bifurcation, perpendicularity, grid formation, polygonal expansion, overlapping circle construction, solids, and higher-dimensional nesting.

1. Core Correction: Do Not Start With the Completed Grid

A mature registry must not be mistaken for a primitive. The square grid, Cartesian axes, sign symmetry, quadrants, and inside/outside partition are powerful but downstream products of construction.

The primitive beginning is simpler:

- one marked referent,
- one possible expression,
- one degree of freedom.

At the root, there is no x/y, no horizontal/vertical, no positive/negative, no square, no triangle, no circle lattice, and no global inside/outside. There is only a distinguishable mark with a possible expression. Every later registry is either imposed directly or constructed from an imposed frame of consideration.

Protocol rule: do not smuggle a mature coordinate system into the primitive.

2. Root Primitive

The root condition is:

MARK + ONE DEGREE OF FREEDOM

A mark is a distinguishable referent. A degree of freedom is the possibility of expression from that referent. Expression produces trace. Repeated or stabilized trace produces axial relation.

Primitive notation:

R0 = marked referent

D1 = one admissible degree of freedom

E1 = expression along D1

R0 --E1--> R1

The axis is not prior to expression. It is inferred from expression.

3. Primitive Ladder of Construction

The ladder should be represented as constructed, not assumed:

1. Mark: a distinguishable referent appears.
2. One degree of freedom: the mark may express.
3. Trace: expression leaves a path.
4. Axis: repeated expression stabilizes as axial relation.
5. Reciprocity: the trace can be traversed outward and inward.
6. Doubling/halving: outward enumeration and inward division appear as a harmonic ladder.
7. Chirality: expression can turn.
8. Closure: turning permits return.
9. Triangle: stepwise closure produces minimal enclosure.
10. Circle: continuous turning produces smooth return.
11. Bifurcation: a second expression distinguishes itself from the first.

12. Perpendicularity: the second expression stabilizes as orthogonal contrast.
13. Square grid: equalized perpendicular axes produce coordinate registry.
14. Polygonal expansion: repeated stepwise turns produce higher polygons.
15. Overlapping circle lattice: radius transfer produces neighbor-node resonance.
16. Solid: lifted closure produces volumetric registry.
17. Higher-dimensional nesting: additional independent degrees, fibers, phases, or lifted relations extend the registry ladder.

Compact maxim:

One degree of freedom first. Axis second. Turn third. Closure fourth. Grid later.

The square does not found the ladder. The ladder constructs the square.

4. L-Frame and C-Frame

The primitive complement is not square versus circle. The more basic distinction is linear expression versus chiral turning.

L-frame: linear extension, axial registry, displacement, count, step, distinction.

C-frame: chiral turning, return, curvature, phase, closure, resonance.

Line gives distance. Chirality gives return.

If linear expression and chiral turning occur simultaneously, the trace becomes spiral or helical. If they occur stepwise, the trace becomes polygonal. If turning continues without stopping, the trace becomes curved or circular.

No chirality, no closure. No closure, no enclosure. No enclosure, no inside/outside.

5. Euclidean / Square Registry With Axis Mundi

The Euclidean or square grid should be treated as an axis-mundi registry constructed from primitive expression.

Construction path:

mark -> one degree of freedom -> axial expression -> reciprocal traversal -> bifurcation -> perpendicular stabilization -> balanced extension -> square / Cartesian registry

Only after this construction do the familiar terms become available:

- horizontal / vertical,
- x / y,
- positive / negative,
- quadrants,
- coordinate address,

- rectangular containment,
- inside / outside by coordinate partition.

These are not primitive terms. They are mature affordances of the square registry.

The square registry is best described as:

axial expression + perpendicular bifurcation + equalized bilateral measure

It excels at coordinate checking, orthogonal projection, sign symmetry, rectangular containment, and Boolean-style partition. It risks flattening curvature, phase, and spectral relation into binary or coordinate forms before those frames have been declared.

6. Triangle Registry and Triangular-Number Progression

The triangle registry should not be treated as a square grid with diagonals added. It encodes a different construction.

The triangle is the first stepwise polygonal closure. A line cannot enclose. Two lines do not enclose. Three segments can produce the first enclosure.

Triangle = minimal closure / triadic support / first simplex relation

The triangular grid progresses through triangular numbers. Its center is not the static Euclidean center of a square grid. Its center can appear occupied and vacant intermittently as the triangular-number construction advances. The construction re-centers or redistributes relation as the triangular field grows.

This gives the triangle registry distinct affordances:

- triadic relation,
- 60-degree adjacency,
- 120-degree recurrence,
- minimal enclosure,
- recursive subdivision,
- local mesh stability,
- intermittent center occupancy,
- simplex-style support.

The triangle stores relation as mutual support among three anchors, not primarily as bilateral opposition.

7. Circle Registry as Sacred-Geometry / Neighbor-Circle Lattice

The circle registry intended here is not a set of concentric rings. It is a compass construction.

Construction path:

18. Draw a circle around origin at any chosen scale. The chosen radius functions as an invariant unit for that construction.
19. Choose a point on the circumference.
20. Draw the next circle from that surface point using the same radius.
21. Let centers, intersections, and overlap lenses become new construction points.
22. Continue by same-radius neighbor-circle expansion.

Thus the circle grid is an overlapping neighbor-circle lattice, like sacred-geometry or flower-of-life construction. It grows by radius steps and local overlap, not by nested shells circumscribing one another.

Circle registry affordances:

- radius preservation,
- chiral return,
- phase relation,
- vesica / overlap lens,
- intersection nodes,
- neighbor transfer,
- inscription and circumscription,
- scale translation,
- polygon and solid projection from centers and overlap points.

This C-frame registry translates line measure into curved return without claiming that line has captured the circle. It gives a constructive lattice where resonance nodes can be mapped between linear standard and spectral continuum.

8. Hybrid Registry: Square Checks, Triangle Closes, Circle Returns

A useful primitive map does not marry one registry. It uses the differences between registries and records transfer cost.

Square checks: coordinate evidence, perpendicularity, bilateral measure, orthogonal verification.

Triangle closes: minimal enclosure, triadic support, simplex propagation.

Circle returns: phase, chirality, radius transfer, continuous closure, overlap-node resonance.

The hybrid registry becomes powerful where these registries agree without being collapsed into literal sameness. Agreement nodes may include:

- circle intersections that land on grid points,
- polygon vertices aligned with circle-overlap nodes,
- triangle subdivisions sharing square-coordinate addresses,
- radii that constructively inscribe or circumscribe polygonal forms,
- modular cycles that complete shared-factor or coprime return paths.

Such nodes act as registry-transfer points. They are not universal truth points. They are low-residue transfer channels between declared registries.

9. Registry Choice as Evidence Grammar

A stimulus field does not arrive with one universal accounting attached. An observer, sensor, culture, or formal system stores it through a registry. The registry decides what counts as evidence.

Square evidence:

coordinate agreement, orthogonal closure, signed extension, rectangular containment.

Triangle evidence:

minimal closure, triadic support, recursive subdivision, local mesh stability.

Circle evidence:

phase relation, overlap node, radius transfer, return, harmonic resonance.

Hybrid evidence:

a relation survives transfer between registries with declared residue.

Therefore:

Different registry. Different admissible evidence. Different search landscape. Different compression. Different residue.

10. Framing, Construction, and Infinite Regress

All framings are either imposed or constructed. If constructed, they are constructed from a prior imposed frame of consideration. This exposes an infinite regress if any one frame is treated as final ontology.

The DST correction is not to deny frames. It is to declare them.

A registry is a fidelity contract:

- this is what this registry can mark,
- this is what it can compare,
- this is what it can compress,
- this is what it can transfer,
- this is what it leaves as residue.

A registry should not be released as global fidelity. It should be released as scoped fidelity under declared construction.

11. Prime / Base / P-Adic Bridge

Prime artifacts provide a mathematical foothold for the registry principle.

Primality is not created by a register. It is an irreducibility relation under factor reproduction. But prime artifacts appear differently depending on the registry through which divisibility is measured and displayed.

The p-adic frame helps explain this for a mathematically trained reader. In an ordinary metric, numbers are close when their difference is small in magnitude. In a p-adic metric, numbers are close when their difference is deeply divisible by a chosen prime p . This proves that divisibility-depth can function as a metric of nearness, depth, and scale.

A base is not only notation. A base can act as a divided measuring whole. Its primitive divisors define native resonance paths. Its residues define visible artifacts.

Counting can be read reciprocally:

- outward enumeration,
- inward division,
- doubling,
- halving,
- modular return,
- residue display.

Primality appears as the irreducible residue of the counting ladder when outward enumeration is continually tested against inward divisibility.

12. Color, Metrology, and Sensing Registries

Color gives an everyday example of registry-dependent legibility.

A continuous spectrum does not arrive already divided into universal color names. It becomes color through sensor bands, neural compression, cultural naming, instrument standards, display gamuts, and measurement conventions.

Color is light registered through a sensing taxonomy.

Likewise, prime artifacts are divisibility registered through base/modular measuring wholes. Harmonic nodes are resonance registered through grids, chambers, and sampling frames.

This does not make color, number, or measurement unreal. It makes them registry-real: real as stored, compared, named, measured, and reused inside a declared observer/instrument frame.

13. PvNP Application

The PvNP game should model the primitive experience of search/verification asymmetry rather than claim to solve the formal P vs NP problem.

P-like condition:

a proposed relation can be checked efficiently inside the selected registry once the certificate and frame are declared.

NP-like condition:

a relation may be globally witnessable or verifiable once found, while finding the correct registry path, closure node, or transfer point may require high search cost.

The game question becomes:

- What registry makes the relation visible?
- What registry makes the certificate checkable?
- What transfer path connects the search frame to the verification frame?
- What residue appears when the wrong registry is used?

The game should therefore model:

- primitive expression,
- registry construction,
- search through possible registry paths,
- certificate checking once a path is declared,
- transfer residue between registries.

14. Implementation Specification for PvNP Simulation

Registry selection should affect dynamics, not merely visual background.

Square mode should alter:

movement snap tendency, orthogonal resistance, axis-aligned verification bonus, perpendicular tension gate, rectangular containment checks.

Triangle mode should alter:

60-degree turn preference, triadic propagation, simplex closure detection, local cluster resonance, recursive subdivision paths, center occupancy/vacancy state.

Circle-overlap mode should alter:

phase accumulation, same-radius neighbor-circle construction, overlap-node attraction, radius-scaled transfer, chiral memory, resonance-node detection.

Hybrid mode should detect:

coincidence between square nodes, triangle nodes, circle centers, and circle-intersection nodes.

Those coincidence points become high-value registry-transfer nodes.

15. Protocol Block: Primitive Registry Analysis

Use this whenever applying the primitive registry ladder.

23. Mark: What is the first distinguishable referent?
24. Degree of freedom: What is the first possible expression?
25. Trace: What axis or path is produced by expression?
26. Return: Can the expression reverse, recur, or close?
27. Chirality: Where does turning enter?
28. Closure: Does closure occur as triangle, circle, polygon, spiral, or lifted form?
29. Bifurcation: What second expression appears?
30. Orthogonality: Is perpendicularity constructed or merely assumed?
31. Registry: Which mature registry is being used: square, triangle, circle-overlap, or hybrid?
32. Evidence: What counts as evidence inside that registry?
33. Transfer: Can the relation transfer to another registry?
34. Residue: What does not survive transfer?
35. Release status: formal / model / heuristic / analogical / semantic / speculative / HELD.

16. DST Release Hygiene

This protocol should preserve the primitive without overstating the bridge.

Do not say: this shape literally is the same structure everywhere.

Say: this primitive relation-function leaves a recognizable signature across registries.

Do not say: one registry proves global truth.

Say: one registry makes certain relations legible and leaves others as residue.

Do not say: P vs NP is solved by grid choice.

Say: the PvNP game models search/verification asymmetry as registry-dependent legibility and transfer cost.

Do not say: primes are created by base.

Say: primality is a factor-reproduction irreducibility relation; base/modular registries reveal different prime artifacts.

Do not say: color is merely subjective illusion.

Say: color is spectrum registered through a sensing and naming taxonomy.

17. Formal Synthesis Statement

A primitive registry system begins with one mark and one possible expression. Repeated expression produces trace and then axis. Return introduces reciprocity. Turning introduces chirality. Stepwise turning produces polygonal closure; continuous turning produces circular closure. Bifurcation introduces a second expression, and only after that expression stabilizes as perpendicular does square-grid logic become available. Thus horizontal/vertical, positive/negative, x/y, inside/outside, and quadrants are not primitive givens; they are downstream products of a constructed registry.

The triangle grid arises from minimal closure and triangular-number progression, with center occupancy shifting as the construction advances. The circle grid arises from sacred-geometry compass construction: same-radius neighbor circles drawn from surface points, producing overlap

lenses, centers, and intersections from which polygons and solids can be projected. These registries do not simply relabel the same content. They change the grammar of admissible evidence and therefore alter what appears compressible, stable, searchable, or verifiable.

All framings are imposed or constructed from prior impositions. A high-fidelity semantic protocol does not deny this regress; it records it. Each registry is released as a scoped fidelity contract, not universal accounting. The task is to keep the primitive marker visible, trace the operation, watch the invariant, record residue, and transfer only what survives.

18. Compact Maxims

One degree of freedom first.

Axis second.

Turn third.

Closure fourth.

Grid later.

The square does not found the ladder.

The ladder constructs the square.

Square checks.

Triangle closes.

Circle returns.

The stimulus does not dictate one universal map.

The observer stores stimulus through a registry.

The registry determines legibility.

Legibility determines evidence.

Evidence determines what can be checked.

What cannot transfer without loss becomes residue.

Same primitive question.

Different registry.

Different artifact.

No forced literal sameness.

No loss of semantic continuity.

19. Landscape Cartographer Compatibility

This module should be treated as canon-adjacent unless formally absorbed. Its proper Landscape Cartographer status is HELD-TOWARD-LOAD-BEARING.

Holder declaration:

The holder is the Primitive Semantic Registry / METALABE / DST ecosystem, using the Landscape Cartographer distinction between LOAD-BEARING, HELD, REDUNDANT, CONCLUDED, and DISCARDED.

Purpose:

To provide a system-readable grammar for translating external structures into primitive registry terms without erasing frame, evidence grammar, or residue.

Bridge toward LOAD-BEARING:

The module becomes load-bearing when it consistently improves DST intake, semantic mapping, PvNP simulation design, and cross-domain translation without increasing overclaim or category conflation.

Current HELD residue:

Exact formalization of transfer cost between square, triangle, and circle-overlap registries remains open. Simulation rules for center occupancy in triangular-number progression and circle-lattice node transfer require implementation and testing.